

## 1. Arithmetic Intensity

Consider the multiplication of dense matrices:

$$C \leftarrow A \cdot B, \tag{1}$$

where  $A$ ,  $B$ , and  $C$  are square matrices of size  $N$ . The elements of the matrices are double-precision floating-point numbers.

Compute the information exchange functions  $I_{\text{fp}}$  and  $I_{\text{mem}}$  as well as the arithmetic intensity AI as a function of  $N$ . Document any assumptions, which you make, about exploitation of data locality.

## 2. Semi-Analytical Performance Modelling and the Roofline Model

Repeat the semi-analytical modelling approach, which has been shown in the lecture, for matrix-matrix multiplication, i.e. the numerical tasks shown in Eq. (1) on a system of your choice. Provide the following documentation:

- a) Central values and variability of the results as a function of  $N$
- b) Benchmarked system
- c) Benchmark code

Furthermore, create a roofline model for your system and compare attained performance with attainable performance. Some hints in this context:

1. Typically, information about the installed memory is available in the BIOS. On a Linux system, the command `dmidecode` can be used to read the BIOS. See, “dmidecode command in Linux with Examples” (GeeksforGeeks, 2024) for more information. Note that `dmidecode` requires root privileges.
2. Each Kunpeng processor on the HAICGU cluster features 8 memory channels populated with DDR4-2666 memory resulting in a theoretical peak bandwidth of 170.6 GB/s per socket.

## 3. Queueing Theory

Assume a memory system with the following attributes:

- Minimal residence time  $R = 400 \text{ ns}$  for loading a cache line (CL)
- Maximum throughput  $X = 0.4 \cdot 10^9 \text{ CL/s}$

How many requests have to be in the system to reach maximum bandwidth?